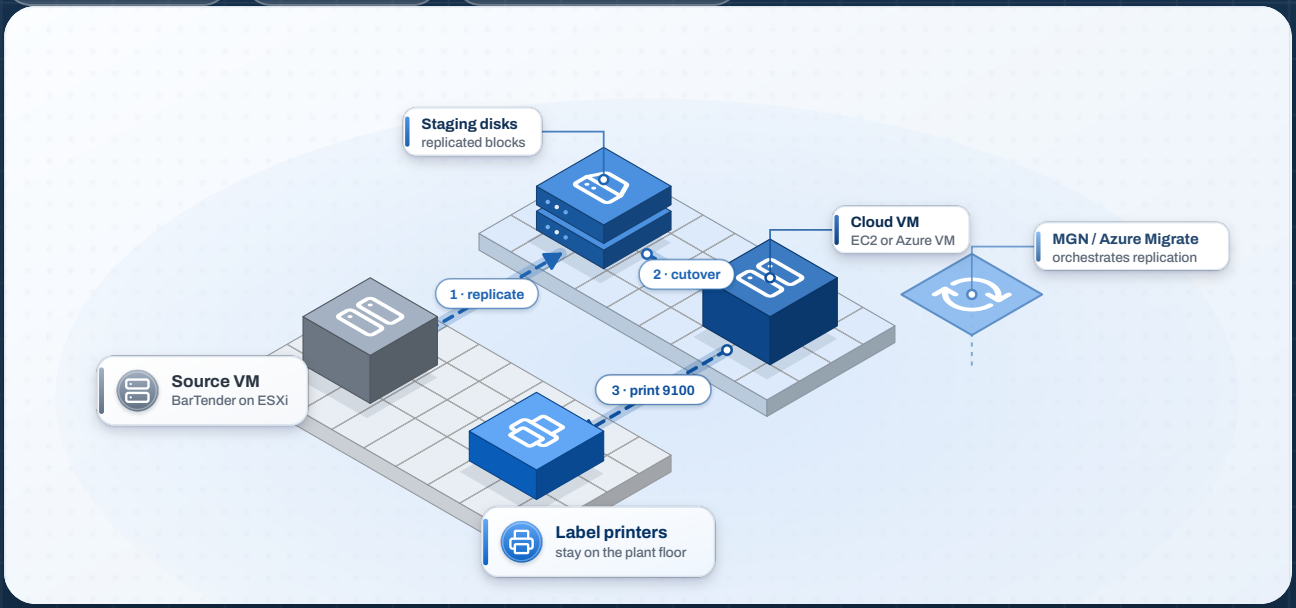
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-MIG-001

Lift-and-Shift Migration of a Label Printing Server to the Cloud

Rehost a BarTender label printing server with AWS Application Migration Service or Azure Migrate, or rebuild it fresh, with tested cutover, printer connectivity and rollback.

Cloud Migration | AWS & Azure | Change Management



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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Place a server in the 7 Rs of migration and justify rehost versus replatform for a label printing system.
- Explain block-level replication and the difference between a test launch and a cutover.
- Rehost a Windows server with AWS Application Migration Service (MGN) and with Azure Migrate, in the console and with scripts.
- Plan printer connectivity, licensing and hard-coded dependencies so labels print correctly from the cloud.
- Run a time-boxed cutover with a validated rollback path and evidence mapped to NIST SP 800-53 controls.
- Assign least-privilege AWS IAM policies and Azure RBAC roles for each migration task.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
2–3 weeks elapsed; 4-hour cutover window	Advanced	MGN full access (AWS) or Contributor on migration RGs (Azure)	Medium: label printing outage if cutover fails

CHAPTER 1

Concepts You Need First

A label printing server looks small: one Windows VM running BarTender, a handful of templates and a few printers. It is also one of the least forgiving systems to move. Labels on regulated products must print exactly as validated, from the right template, on the right printer, and with a valid licence. This SOP moves such a server from an on-premises VMware host into a public cloud with minimum downtime, and teaches the concepts that make the move predictable.

1.1 The 7 Rs of migration

Every workload in a cloud programme is assigned a *migration strategy*. The industry vocabulary, popularised by AWS and used by Microsoft's Cloud Adoption Framework in similar form, is the "7 Rs":

Strategy	Meaning	Applied to a label server
Retire	Switch it off; nobody needs it.	Only if labels are produced elsewhere.
Retain	Leave it on-premises for now.	If printers and latency make cloud printing impractical.
Rehost	"Lift and shift" the server unchanged onto cloud VMs.	**Methods A and B.** Fastest, lowest validation impact.
Relocate	Move the hypervisor layer itself (for example VMware Cloud on AWS, Azure VMware Solution).	Only when a whole VMware estate moves.
Replatform	Small optimisations during the move: new OS version, managed database.	**Method C** (fresh VM + reinstall).
Repurchase	Replace with SaaS.	A cloud labeling service, as a later project.
Refactor	Re-architect for cloud-native services.	Rarely justified for a print server.

KEY CONCEPT · WHY REHOST IS THE DEFAULT FOR VALIDATED SYSTEMS

A rehost copies the disks block for block. The OS, application binaries, registry, templates and configuration are identical, so the validated state is preserved and re-testing can focus on what actually changed: the network path, the licence and the hardware the OS sees. A rebuild changes everything at once and usually needs a fuller re-validation.

1.2 Block-level replication

Both AWS Application Migration Service (MGN) and Azure Migrate rehost by **replicating disk blocks**, not files. Think of the source disk as a large grid of numbered boxes. The tool first copies every box (the *initial sync*), then watches which boxes change and sends only those. Because it works below the file system, it does not care whether the change is a database page, a template or a registry hive, and the application keeps running during replication.

- **AWS MGN** installs the AWS Replication Agent in the source OS. The agent streams changed blocks continuously over TCP 1500 to lightweight replication servers in a staging subnet, which write them to EBS volumes. Recovery points are kept as EBS snapshots.
- **Azure Migrate** offers *agentless* replication for VMware (the Azure Migrate appliance uses vSphere snapshots and changed block tracking, sending deltas in cycles) and *agent-based* replication (Mobility service in the guest plus a replication appliance), which also covers physical servers and other clouds.

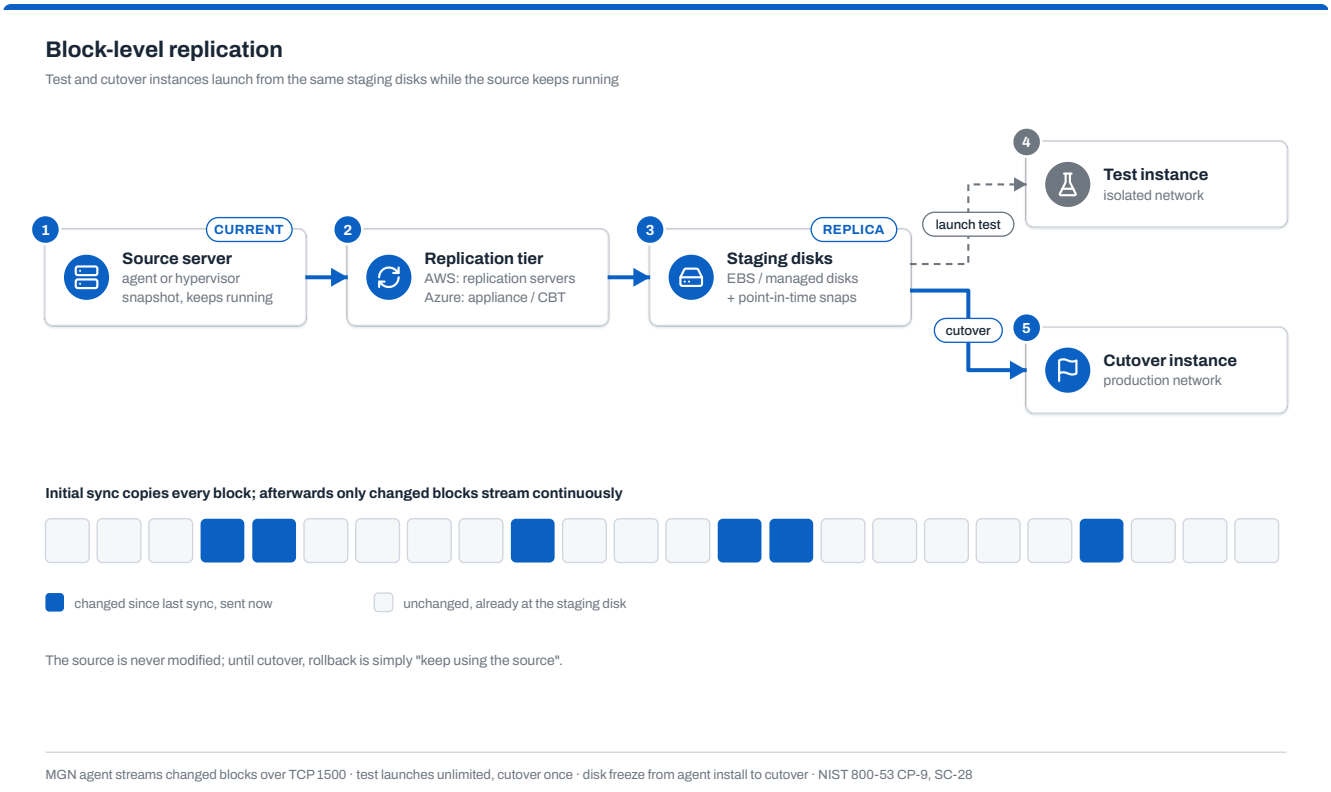


Figure 1. Because every launch comes from the replicated staging disks, you can test as often as needed without touching the source, and cutover is just the last launch.

1.3 Test launch versus cutover

A **test launch** boots a copy of the replicated server in an isolated network while replication continues. Nothing changes for users. It proves the server boots, drivers load and the application starts. A **cutover** is the real move: users stop working, the final changes replicate, the cloud instance becomes production and the source is shut down. Because replication continues after a test, you can test as many times as needed; cutover happens once, inside an approved window.

WHY IT MATTERS · THE DISK-CHANGE FREEZE

After the agent is installed, adding, removing or resizing source disks breaks or restarts replication. Treat agent installation as the start of a change freeze for the server's disk layout; any exception goes back through the change advisory board.

1.4 Things that break when a label server moves

Dependency	Why it breaks	Mitigation
Licensing	BarTender activations are tied to the machine; new virtual hardware (and MAC address) can look like a different computer. Remote clients find the licence through the BarTender Licensing Service (TCP 5160 by default).	Confirm with the vendor/reseller; deactivate before, reactivate after; open 5160 if clients use a remote licence server.
Hard-coded IPs and UNC paths	Templates, Integration Builder, scripts and printer ports may point at `192.0.2.x` or `\\fileserver\labels`.	Discovery script (Chapter 2); replace with DNS names before cutover.
Printer connectivity	Jobs are rendered on the server and sent to printers over TCP 9100 (RAW). From the cloud they cross a VPN or private circuit.	Routes and firewall for 9100 (plus SNMP 161/UDP for status); measure latency; test large jobs.
Identity	Local accounts versus Active Directory; the cloud VM must reach a domain controller.	Test logons on the test instance.
Database	The BarTender System Database (SQL Server) is referenced by name.	Use DNS, not IP; verify connectivity from the cloud subnet.

1.5 Printing from the cloud

When BarTender runs in the cloud, the printers stay on the plant or lab floor. Each print job is generated on the server by the printer driver and sent as a raw byte stream to `printer:9100`. That stream now travels over a site-to-site VPN, AWS Direct Connect or Azure

ExpressRoute. TCP handles latency well, but large graphic-heavy jobs and printers that expect quick bidirectional status replies can slow down. Measure round-trip time from the cloud subnet to each printer during the test launch and print the largest real template, not just a one-line test label.

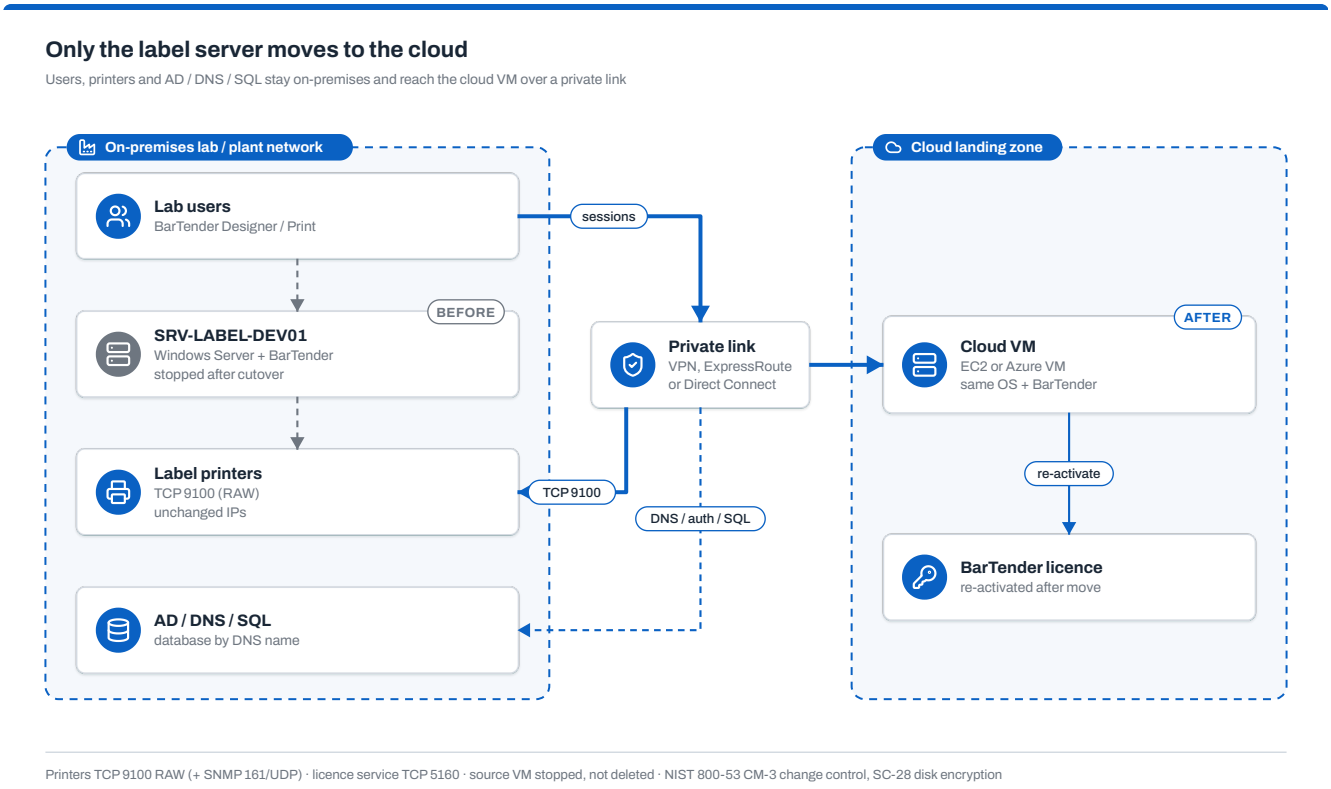


Figure 2. Only the server changes location, so re-testing focuses on what actually moved: the network path to printers and dependencies, and the licence.

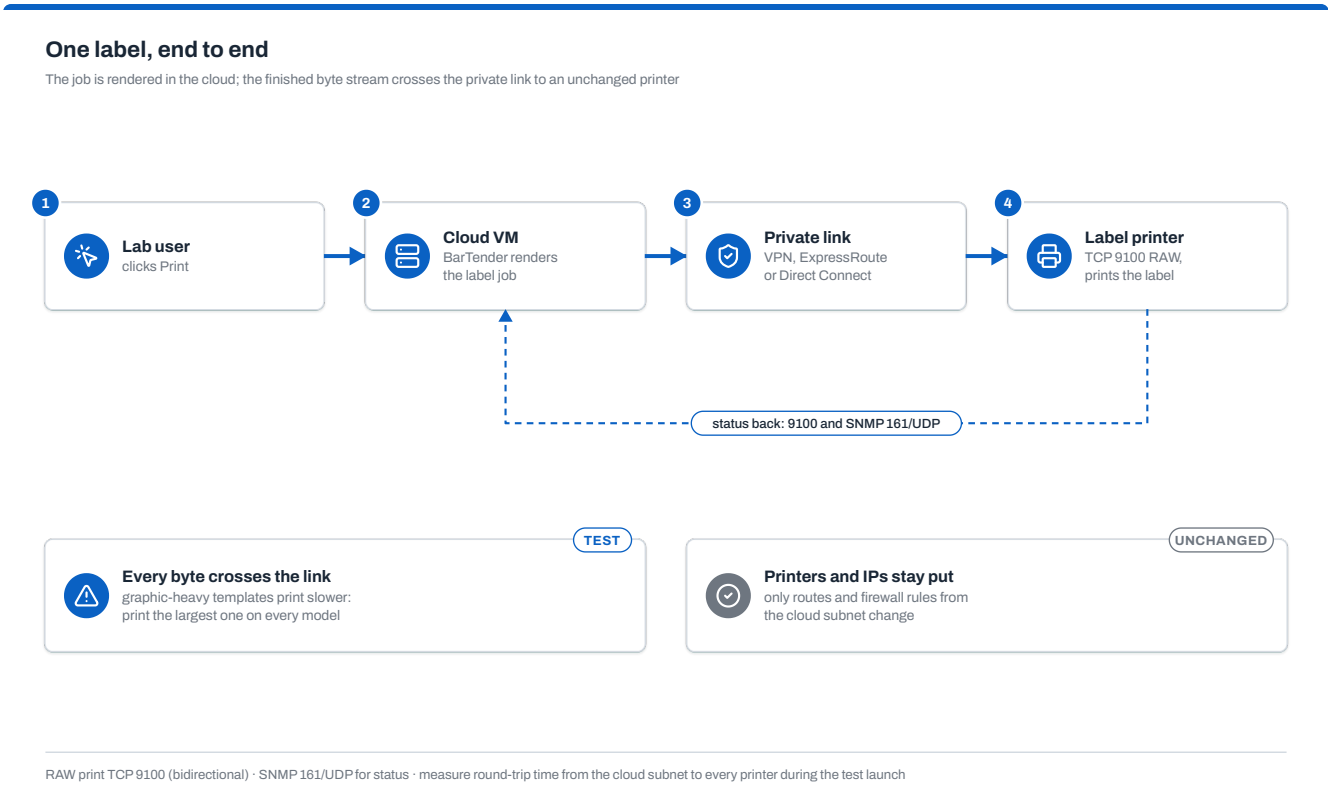


Figure 3. Every byte of every label now crosses the private link, so the test day must print the largest, most graphic-heavy template on every printer model.

1.6 Choosing a method

A · AWS MGN

Agent-based block replication to EC2; console or AWS CLI.

BEST FOR: AWS LANDING ZONES (THE REFERENCE ENGAGEMENT)

B · Azure Migrate

Agentless (VMware) or agent-based replication to Azure VMs; portal or Az.Migrate PowerShell.

BEST FOR: AZURE LANDING ZONES

C · Rebuild / replatform

New VM, fresh OS, reinstall BarTender, migrate templates and database.

BEST FOR: UNSUPPORTED OS, HEAVY DRIFT, VERSION UPGRADE

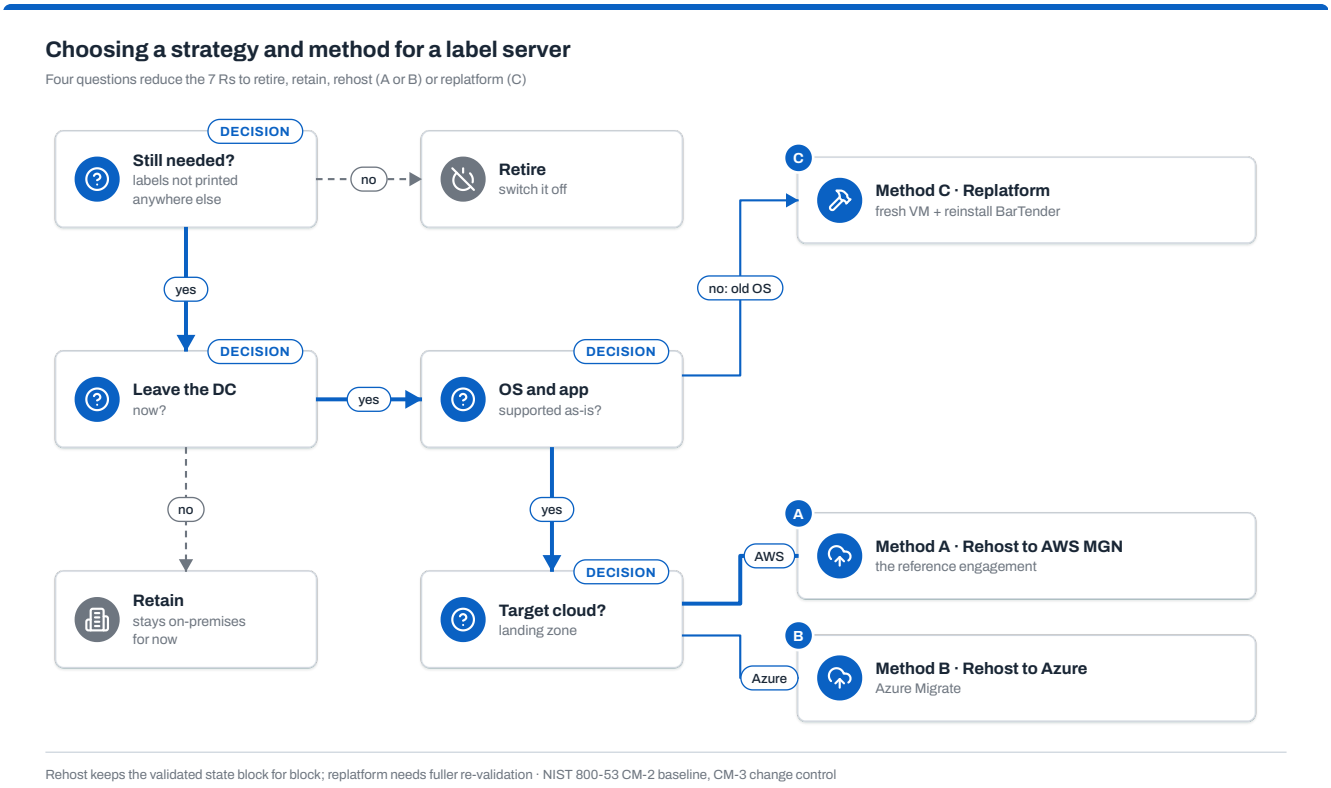


Figure 4. Rehost (Method A or B) is the default for a validated label server because it preserves the validated state; rebuild only when the OS or application cannot move as-is.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Roles and responsibilities

Role	Responsibility
Infrastructure lead	Source VM health, disk layout, firewall and routing changes, VPN or private circuit.
Cloud migration team	Replication agent or appliance, launch settings, test and cutover execution, cutover coordination.
Labeling SME	Template and printer validation, licence checks, post-cutover business sign-off.
Security analyst	Control evidence against NIST SP 800-53 (AC, AU, CM, CP, SC families).
Change advisory board	Approves the agent-install change, the cutover change and any exception to the disk freeze or the non-production schedule.

2.2 Level of access required

Task	Least-privilege role / policy	Scope and notes
Operate MGN (console, CLI)	AWS managed policy `AWSApplicationMigrationFullAccess`	Target account. First-time *initialize service* also creates service-linked roles, so it needs IAM permissions to create them.
Install the AWS Replication Agent	`AWSApplicationMigrationAgentInstallationPolicy`	Short-lived credentials (role session), never a long-term access key left on the server.
Launch and inspect test/cutover EC2	`AWSApplicationMigrationEC2Access` or scoped EC2 permissions	Target VPC and subnets only.
Audit AWS migration	`AWSApplicationMigrationReadOnlyAccess`	Plus CloudTrail read for evidence.

Create Azure Migrate project and register appliance	Owner, or Contributor + User Access Administrator	Subscription or project RG, first time only (the appliance's Entra apps receive role assignments). Application Developer in Entra ID if app registration is restricted.
Replicate, test and migrate in Azure	Contributor	Project RG and target RG.
Attach NICs to target subnet	Network Contributor, or a custom role with `Microsoft.Network/virtualNetworks/subnets/join/action`	Target VNet.
vCenter access for agentless discovery and replication	vCenter role with read, snapshot, disk access and change-tracking privileges	Only the VMs in scope.
Source server	Local Administrator	Agent install, discovery script, service stop at cutover.
BarTender and database (Method C)	BarTender administrator; SQL `db_owner` (or sysadmin for restore)	Licence activation, System Database backup and restore.

GOOD PRACTICE · JUST-IN-TIME PRIVILEGE

In AWS, grant the policies through IAM Identity Center permission sets with short session durations. In Azure, make Contributor and Owner *eligible* through Microsoft Entra Privileged Identity Management and activate them per change with the change number as justification. Both produce audit evidence for AC-2 and AC-6.

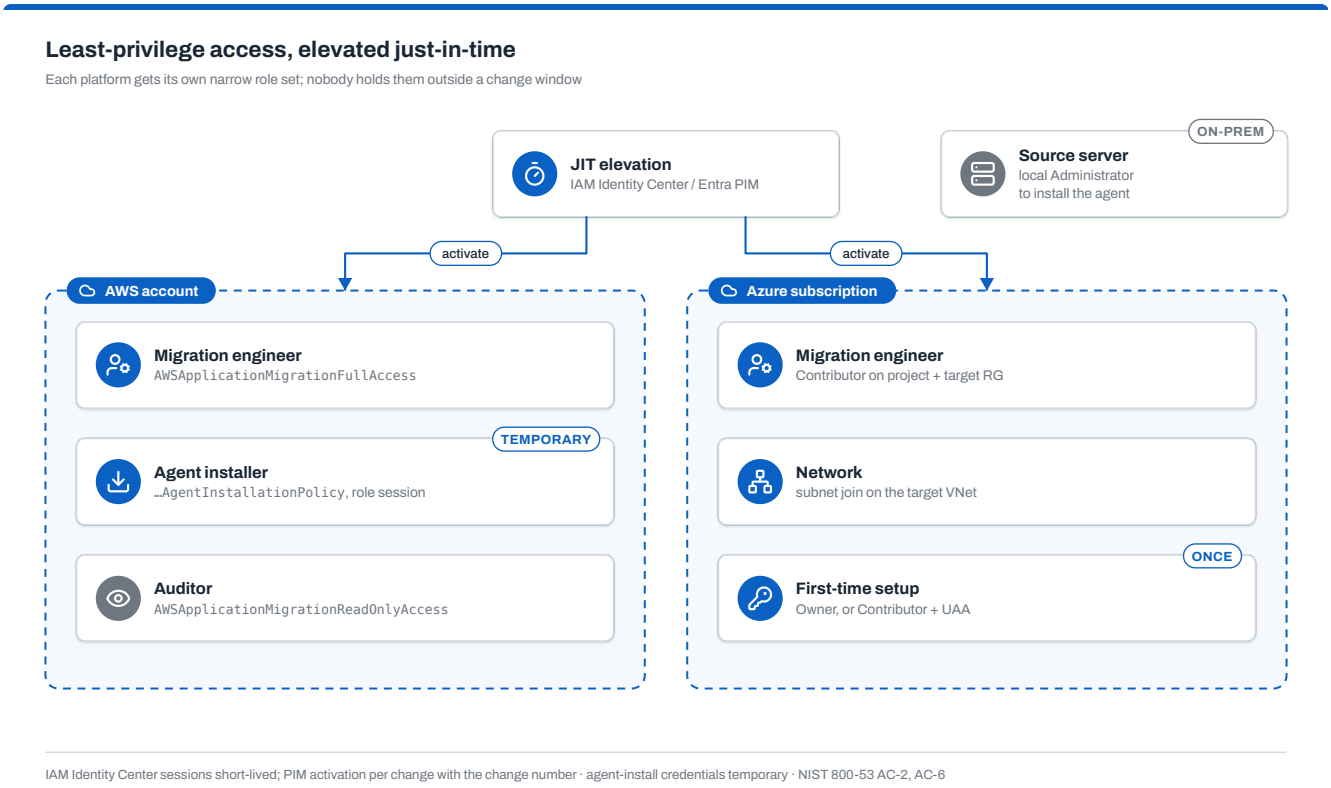


Figure 5. Every migration role is eligible rather than standing and is activated per change, which produces the AC-2 and AC-6 evidence the security analyst needs.

2.3 Pre-migration checklist

- ☐ Architecture review completed: no load balancer, no B2B/EDI connections, no VDI dependency, database referenced by DNS, no on-premises NAS or DR dependency (or each one documented).
- ☐ Right-sizing agreed. In the reference engagement the source ran at about 92% CPU and 85% memory on 2 vCPU / 8 GB, so the target was sized up to 4 vCPU / 16 GB rather than copied.
- ☐ Licensing model confirmed with the vendor for the new platform; deactivation and reactivation steps written down.
- ☐ Hard-coded IPs, UNC paths and printer ports discovered and remediated (script below).
- ☐ Template and log locations confirmed and archived; a full backup or VM snapshot taken before agent install.
- ☐ Private connectivity (VPN, Direct Connect or ExpressRoute) carries DNS, AD, SQL and TCP 9100 to every printer subnet.
- ☐ Change 1 (agent install) and Change 2 (cutover) raised; cutover window agreed with stakeholders; SME booked for one full business day of testing.
- ☐ Non-production schedule agreed: dev and test instances stopped off-hours (in the reference programme about 79 of 168 hours per week), with opt-out by justified

exception only.

2.4 Discovery script: hard-coded dependencies

Run this on the source server before replication starts. It inventories printers and ports, mapped drives, services and scheduled tasks, and searches BarTender-related configuration for literal IP addresses and UNC paths, writing everything to a JSON evidence file.

GET-LABELSERVERDEPENDENCIES.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Inventory dependencies of a label printing server before migration (read-only).
.NOTES     Run elevated on the source server. Output: JSON evidence file for the change record.
#>
[CmdletBinding()]
param(
  [string[]] $SearchPaths = @("$env:ProgramData\Seagull", "$env:ProgramFiles\Seagull", 'D:\Labels'),  # adjust
  [string[]] $Extensions  = @('*.xml','*.config','*.ini','*.json','*.bat','*.ps1','*.btin'),
  [string]   $OutFile     = ".\$env:COMPUTERNAME-dependencies.json"
)
$ErrorActionPreference = 'Stop'
$ipRx  = '\b(?:\d{1,3}\.){3}\d{1,3}\b'
$uncRx = '\\\\[A-Za-z0-9._-]+\\[^\s"''<>]+'

$hits = foreach ($p in $SearchPaths | Where-Object { Test-Path $_ }) {
  Get-ChildItem -Path $p -Recurse -File -Include $Extensions -ErrorAction SilentlyContinue |
    Select-String -Pattern $ipRx, $uncRx -AllMatches |
    ForEach-Object { [pscustomobject]@{ File = $_.Path; Line = $_.LineNumber; Match = ($_.Matches.Value -join ', ') } }
}
$report = [ordered]@{
  Computer       = $env:COMPUTERNAME; CollectedUtc = (Get-Date).ToUniversalTime().ToString('s')
  OS             = (Get-CimInstance Win32_OperatingSystem | Select-Object Caption, Version)
  Printers       = Get-Printer | Select-Object Name, DriverName, PortName, Shared
  PrinterPorts   = Get-PrinterPort | Select-Object Name, PrinterHostAddress, PortNumber
  MappedDrives   = Get-SmbMapping -ErrorAction SilentlyContinue | Select-Object LocalPath, RemotePath
  Services       = Get-Service | Where-Object DisplayName -match 'BarTender|Seagull|SQL' | Select-Object Name, Status, StartType
  ScheduledTasks = Get-ScheduledTask | Where-Object TaskPath -notlike '\Microsoft\*' | Select-Object TaskName, State
  NetConfig      = Get-NetIPConfiguration | Select-Object InterfaceAlias, IPv4Address, DNSServer
  HardCoded      = $hits
}
$report | ConvertTo-Json -Depth 5 | Out-File $OutFile -Encoding utf8
Write-Host "Wrote $OutFile ($($hits.Count) hard-coded IP/UNC hits)"
```

Printer ports whose `PrinterHostAddress` is an IP address are the most common finding. Re-point them to DNS names (for example `prn-line1.contoso.com`) before cutover, so that the cloud server reaches the same printers by name.

CHAPTER 3

Method A · AWS Application Migration Service (MGN)

This is the method used in the reference engagement: a Windows Server 2019 VM on VMware ESXi rehosted to EC2 in a private subnet. The original planning diagram, sanitized, shows the shape of the target.

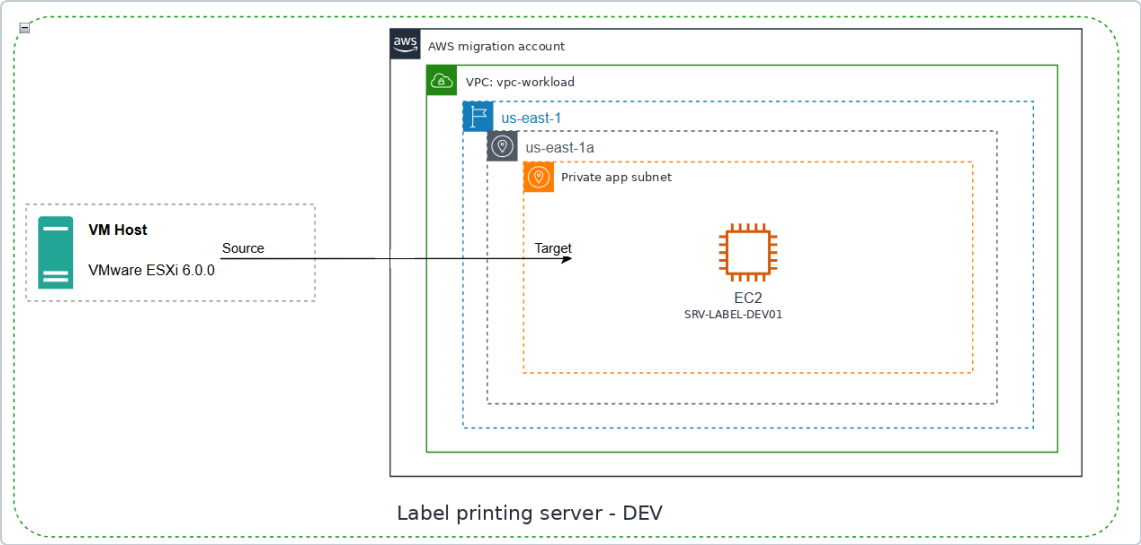


Figure 6. The reference engagement's target architecture: source VM on ESXi replicated into a private application subnet of a workload VPC in us-east-1.

STEP 1 Prepare MGN in the target account

1. Sign in to the AWS console in the target Region and open **Application Migration Service**. On first use choose **Get started** / **Set up service** to initialize it; this creates the service-linked and replication roles.
2. Open **Settings › Replication template**. Select the *staging area subnet*, the replication server instance type, EBS volume type and encryption key. Turn on **Use private IP for data replication** so data flows over the VPN or Direct Connect rather than the internet.
3. Confirm the staging subnet's security group allows TCP 1500 inbound from the source, and that the source can reach the MGN endpoint and Amazon S3 on TCP 443.

STEP 2 Install the replication agent (change 1)

1. Obtain short-lived credentials for a role holding `AWSApplicationMigrationAgentInstallationPolicy`.
2. On the source server, download the Windows installer for your Region from the MGN console (**Add source servers** shows the exact command) and run it elevated:

SOURCE SERVER (ELEVATED) CMD

```
AwsReplicationWindowsInstaller.exe --region us-east-1 ^
  --aws-access-key-id <temporary-key-id> --aws-secret-access-key <temporary-secret> ^
  --aws-session-token <session-token>
```

3. Choose to replicate all disks. The server appears under **Source servers**; initial sync starts and the disk-change freeze begins.
4. Wait for *Ready for testing*. Initial sync time depends on used disk space and bandwidth; after that, replication lag should stay in seconds.

STEP 3 Launch settings and test launch

1. Open the source server, then **Launch settings**. Set *Instance type right sizing* to **Off** and edit the EC2 launch template to the agreed size (for example `m5a.xlarge` for 4 vCPU / 16 GB), the private subnet, security groups and tags. Choose *BYOL* or *license-included* for Windows.
2. Select **Test and cutover › Launch test instances**. Use a test subnet that cannot reach production printers, or tell the SME which printer to use.
3. Connect through Session Manager or a bastion and run the validation list (Chapter 6). Expect BarTender to report a licence change.
4. When satisfied, choose **Mark as "Ready for cutover"**; this terminates the test instance. Replication continues.

STEP 4 Cutover (change 2)

1. At window start, stop the BarTender services on the source, wait until the console shows no replication lag, then choose **Launch cutover instances**.

2. Shut down the source VM (keep it; it is the rollback). Update DNS so the server name resolves to the EC2 private IP, re-activate the licence, and run SME validation.
3. On success choose **Finalize cutover**; this stops replication and removes the replication resources. Later, **Mark as archived** and decommission the source.

3.1 AWS CLI outline

The same lifecycle can be scripted with the `aws mgn` commands (AWS CLI v2). The equivalent PowerShell cmdlets live in the `AWS.Tools.Mgn` module (for example `Start-MGNTest` and `Start-MGNCutover`); list them with `Get-Command -Module AWS.Tools.Mgn`.

MGN-LIFECYCLE.SH · OUTLINE — BASH

```
REGION=us-east-1
aws mgn initialize-service --region $REGION                    # once per account and Region
SID=$(aws mgn describe-source-servers --region $REGION \
      --query "items[?sourceProperties.identificationHints.hostname=='SRV-LABEL-DEV01'].sourceServerID" -
-output text)
aws mgn update-launch-configuration --region $REGION --source-server-id $SID \
      --target-instance-type-right-sizing-method NONE --copy-tags
aws mgn start-test     --region $REGION --source-server-ids $SID          # test launch
aws mgn change-server-life-cycle-state --region $REGION --source-server-id $SID \
      --life-cycle state=READY_FOR_CUTOVER                            # after test sign-off
aws mgn start-cutover  --region $REGION --source-server-ids $SID          # in the window
aws mgn finalize-cutover --region $REGION --source-server-id $SID         # after validation
aws mgn mark-as-archived --region $REGION --source-server-id $SID         # housekeeping
```

NOTE · NON-PRODUCTION SCHEDULING IN AWS

For dev and test instances, the AWS *Instance Scheduler* solution or EventBridge Scheduler with the `AWS-StopEC2Instance` / `AWS-StartEC2Instance` Systems Manager runbooks stop and start instances by tag. A typical 79-hour weekly off-period is nights from 21:00 to 06:00 on weekdays plus most of the weekend.

CHAPTER 4

Method B · Azure Migrate

Azure Migrate is a hub for discovery, assessment and migration. Its **Migration and modernization** tool (formerly Server Migration) replicates servers to Azure VMs. For a VMware VM the recommended path is **agentless** replication through the Azure Migrate appliance; use **agent-based** replication (Mobility service plus a replication appliance) for physical servers, other clouds, or when agentless is not possible.

STEP 1 Create the project and discover the server

1. In the Azure portal open **Azure Migrate › Servers, databases and web apps › Create project**. Choose the subscription, a project resource group (for example `rg-migrate-labels`), a name and a geography.
2. Under **Discover**, generate a project key, download the appliance OVA for VMware, deploy it on vCenter, and register it with the key. Provide a vCenter account with the privileges in Chapter 2.
3. Wait until the label server appears in **Discovered servers**. Optionally create an **Assessment** to get Azure VM size and cost recommendations.

STEP 2 Replicate

1. In **Migration and modernization** select **Replicate**. Source: *Servers or virtual machines (VM)*, *Azure VM*, *VMware vSphere*, and your appliance.
2. **Virtual machines**: select the label server (import sizing from the assessment if you made one).
3. **Target settings**: subscription, target RG (for example `rg-labels-dev`), region, virtual network and subnet, availability option, disk encryption type, and Azure Hybrid Benefit if Windows Server licences with Software Assurance apply.
4. **Compute**: VM name and size (for example `Standard_D4s_v5`); **Disks**: Premium SSD or Standard SSD; **Tags**: ChangeRef, Owner, Environment.
5. **Review + Start replication**. Track progress under **Replicating machines** until initial replication completes and delta cycles run.

STEP 3 **Test migration**

1. In `Replicating machines` open the server and select `Test migration`. Choose an **isolated test VNet**, never the production one.
2. Connect with Azure Bastion and run the validation list. Then select `Clean up test migration` and record the results.

STEP 4 **Migrate (cutover) and complete**

1. In the window, stop BarTender services and select `Migrate`. Choose *Yes* to shut down the source and perform a planned migration with no data loss.
2. Update DNS, re-activate the licence, run SME validation.
3. On success select `Complete migration` (stops replication and removes replication resources). Enable Azure Backup and Update Manager on the new VM.

4.1 Az PowerShell base template

The `Az.Migrate` module exposes the same lifecycle. The template below targets the agentless VMware scenario and runs one stage per invocation, so each stage can be tied to a change step and a log.

INVOKE-AZMIGRATELABELSERVER.PS1 · BASE TEMPLATE POWERSHELL

```powershell
<#
.SYNOPSIS  Rehost one VMware VM to Azure with Azure Migrate (agentless): Replicate | Test | CleanupTest | Migrate | Complete.
.NOTES     Requires Az.Accounts and Az.Migrate. Appliance must already be registered and the VM discovered.
           Roles: Contributor on project RG and target RG; subnet join on the target VNet.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [Parameter(Mandatory)] [string] $SubscriptionId,
  [Parameter(Mandatory)] [string] $ProjectResourceGroup,        # e.g. rg-migrate-labels
  [Parameter(Mandatory)] [string] $ProjectName,
  [Parameter(Mandatory)] [string] $SourceVmName,                # display name in Discovered servers
  [Parameter(Mandatory)] [ValidateSet('Replicate','Test','CleanupTest','Migrate','Complete')] [string] $Stage,
  [Parameter(Mandatory)] [string] $ChangeRef,
  [string] $TargetRegion        = 'eastus2',
  [string] $TargetResourceGroupId,                              # /subscriptions/<sub>/resourceGroups/rg-labels-dev
  [string] $TargetVNetId,                                       # production VNet resource ID
  [string] $TargetSubnetName    = 'snet-app',
  [string] $TestVNetId,                                         # isolated VNet for test migration
```

```
  [string] $TargetVmName        = $SourceVmName,
  [string] $TargetVmSize        = 'Standard_D4s_v5',
  [ValidateSet('Standard_LRS','StandardSSD_LRS','Premium_LRS')] [string] $DiskType = 'Premium_LRS'
)
$ErrorActionPreference = 'Stop'
Connect-AzAccount -Subscription $SubscriptionId | Out-Null

# ---- Locate the discovered server and any existing replication ------------
$server = Get-AzMigrateDiscoveredServer -ProjectName $ProjectName -ResourceGroupName
$ProjectResourceGroup `
            -DisplayName $SourceVmName | Select-Object -First 1
if (-not $server) { throw "Server '$SourceVmName' not found in project $ProjectName." }
$repl = Get-AzMigrateServerReplication -DiscoveredMachineId $server.Id -ErrorAction SilentlyContinue
$job  = $null

switch ($Stage) {
  'Replicate' {
    if ($PSCmdlet.ShouldProcess($SourceVmName, 'Start replication')) {
      # One-time per project and region: sets up storage/Key Vault used by agentless replication
      Initialize-AzMigrateReplicationInfrastructure -ResourceGroupName $ProjectResourceGroup `
          -ProjectName $ProjectName -Scenario agentlessVMware -TargetRegion $TargetRegion
      $job = New-AzMigrateServerReplication -InputObject $server -TargetResourceGroupId
$TargetResourceGroupId `
          -TargetNetworkId $TargetVNetId -TargetSubnetName $TargetSubnetName -TargetVMName $TargetVmName
`
          -TargetVMSize $TargetVmSize -LicenseType WindowsServer -OSDiskID $server.Disk[0].Uuid -DiskType
$DiskType
    }
  }
  'Test' {
    if ($PSCmdlet.ShouldProcess($SourceVmName, 'Test migration into isolated VNet')) {
      $job = Start-AzMigrateTestMigration -InputObject $repl -TestNetworkId $TestVNetId
    }
  }
  'CleanupTest' {
    if ($PSCmdlet.ShouldProcess($SourceVmName, 'Clean up test migration')) {
      $job = Start-AzMigrateTestMigrationCleanup -InputObject $repl
    }
  }
  'Migrate' {
    if ($PSCmdlet.ShouldProcess($SourceVmName, 'Planned migration (shuts down source)')) {
      $job = Start-AzMigrateServerMigration -InputObject $repl -TurnOffSourceServer
    }
  }
  'Complete' {
    if ($PSCmdlet.ShouldProcess($SourceVmName, 'Complete migration (stop replication)')) {
      $job = Remove-AzMigrateServerReplication -InputObject $repl
    }
  }
}

# ---- Wait for the job and write evidence ----------------------------------
if ($job) {
  while ($job.State -in 'InProgress','NotStarted') { Start-Sleep 30; $job = Get-AzMigrateJob -InputObject
$job }
}
$repl = Get-AzMigrateServerReplication -DiscoveredMachineId $server.Id -ErrorAction SilentlyContinue
[pscustomobject]@{
  ChangeRef = $ChangeRef; Stage = $Stage; Operator = (Get-AzContext).Account.Id; TimeUtc = (Get-
Date).ToUniversalTime()
  JobState = $job.State; MigrationState = $repl.MigrationState; StateDescription =
```

```
$repl.MigrationStateDescription
} | ConvertTo-Json | Out-File ".\$ChangeRef-$Stage-evidence.json"
```

EXAMPLE · STAGE BY STAGE

INVOCATION POWERSHELL

```
$common = @{ SubscriptionId='<sub-guid>'; ProjectResourceGroup='rg-migrate-labels';
ProjectName='mig-labels'
            SourceVmName='SRV-LABEL-DEV01'; ChangeRef='CHG0012345' }
.\Invoke-AzMigrateLabelServer.ps1 @common -Stage Replicate -TargetResourceGroupId '<rg-id>' -
TargetVNetId '<vnet-id>' -WhatIf
.\Invoke-AzMigrateLabelServer.ps1 @common -Stage Test -TestVNetId '<test-vnet-id>'
.\Invoke-AzMigrateLabelServer.ps1 @common -Stage CleanupTest
.\Invoke-AzMigrateLabelServer.ps1 @common -Stage Migrate        # inside the cutover window
.\Invoke-AzMigrateLabelServer.ps1 @common -Stage Complete       # after SME sign-off
```

NOTE · AGENT-BASED SCENARIO AND SCHEDULING

For agent-based replication the portal flow is the same after the replication appliance and Mobility service are installed, and the cmdlets take agent-based inputs; check `Get-Help New-AzMigrateServerReplication -Full` for the parameter set your scenario uses. For non-production schedules use *Start/Stop VMs v2* or VM auto-shutdown with an Automation or Logic Apps start schedule.

CHAPTER 5

Method C · Rebuild and Replatform

A rebuild creates a clean VM in the cloud and installs BarTender fresh. It costs more effort and validation, but is the right answer when the source OS is out of support, the server has years of undocumented drift, or a BarTender upgrade is due anyway.

WATCH OUT · VALIDATED OUTPUT

A different driver version, DPI setting or font can shift barcodes or text by a few dots and fail verification. Keep driver versions identical to the source unless the change is part of an approved revalidation.

1. **Build the VM** from a hardened marketplace or gallery image (current Windows Server), in the private subnet, joined to the domain, with backup and patching enabled.
2. **Install prerequisites**: SQL Server (or point at a managed SQL instance), .NET, and the same printer driver versions (Seagull drivers for thermal printers) as the source.
3. **Install BarTender** with the agreed edition and version. Deactivate the licence on the old server first, then activate on the new one through the BarTender licensing wizard or Administration Console.
4. **Migrate the System Database**: back up the BarTender System Database on the source SQL instance and restore it on the target, or use the BarTender Administration Console's database tools to connect to the moved database.
5. **Copy templates and integrations**: label documents (`.btw`), Librarian content, Integration Builder files, scripts and print-only folders, preserving NTFS permissions (`robocopy /MIR /COPYALL`).
6. **Recreate printers** with TCP/IP ports that use DNS names, set the same driver settings (darkness, speed, media), and export/import printer settings where the driver supports it.
7. **Validate** with the full SME test list and compare printed output side by side with labels from the old server.

RECREATE PRINTER PORTS BY DNS NAME (EXAMPLE) POWERSHELL

```
Add-PrinterPort -Name 'IP_prn-line1' -PrinterHostAddress 'prn-line1.contoso.com' -PortNumber 9100
Add-Printer     -Name 'LINE1-ZT411' -DriverName '<exact driver name from source>' -PortName 'IP_prn-
line1'
```

CHAPTER 6

Cutover, Validation and Rollback

6.1 Timeline

Two changes bracket the migration: the agent install (which starts replication and the disk freeze) and the cutover. Between them, test as often as needed. The cutover itself is a short, rehearsed sequence with a hard decision point that leaves time to roll back inside the window.

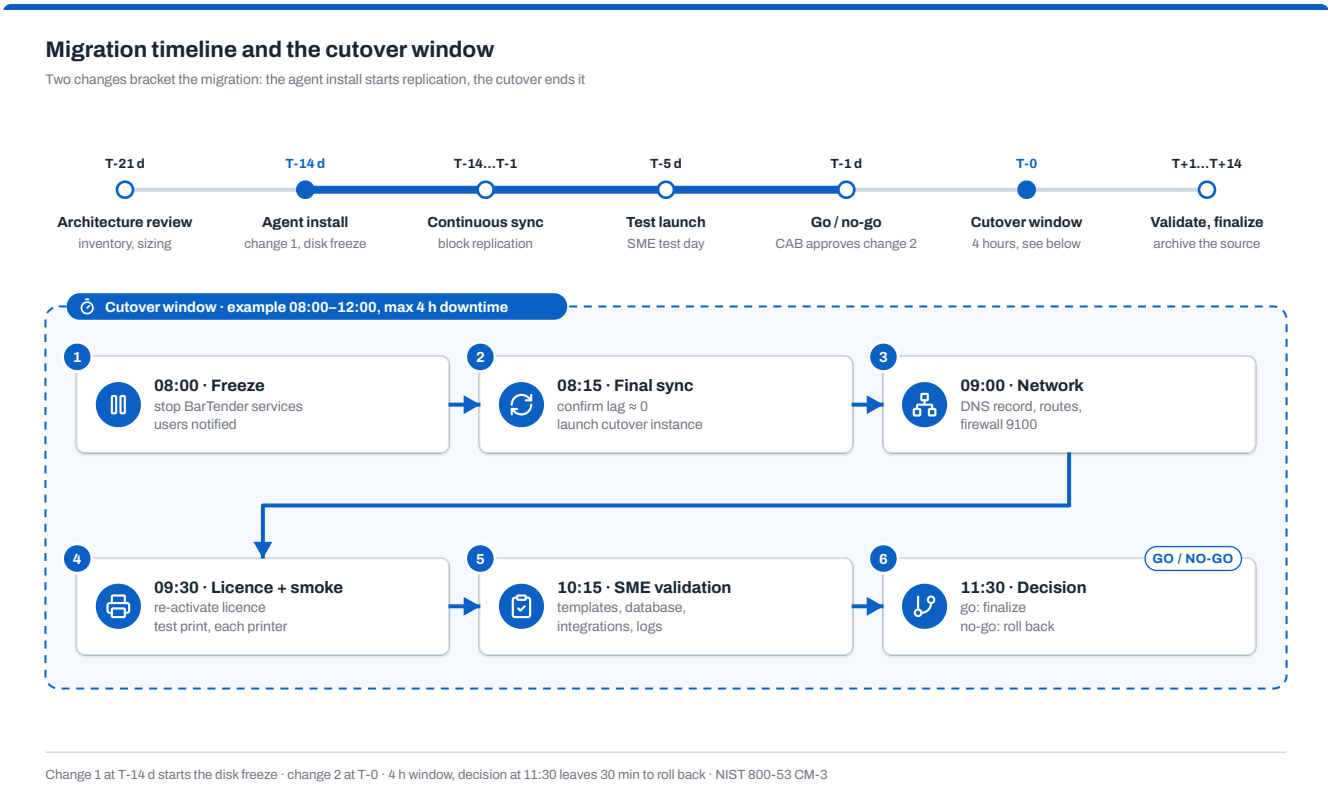


Figure 7. Non-production cutovers can run in business hours; production label servers usually need an evening or weekend window, with the decision point early enough to roll back inside it.

6.2 Cutover steps

1. **Freeze**: announce the outage, stop BarTender services (and the Integration service) on the source so no new jobs or template edits occur.
2. **Final sync and launch**: confirm zero lag (MGN) or run the planned migration (Azure), then launch the cutover instance.
3. **Network**: update the DNS A record for the server name to the cloud private IP (lower its TTL a day earlier), confirm routes and firewall rules for TCP 9100 to each

printer subnet, and SQL and AD reachability.

4. **Licence and smoke test**: re-activate the BarTender licence, start services, print a test label on each production printer.
5. **SME validation**: run the validation list below and record evidence.
6. **Decision**: by the agreed time, either finalize/complete the migration or start rollback.

6.3 Validation list

Functional (Labeling SME)

- [ ] Labels print from dev, test and prod on every printer
- [ ] Largest template prints at normal speed
- [ ] Templates and Librarian files open; revisions intact
- [ ] Integration triggers fire (file drop, web service)
- [ ] Environment separation (dev/test/prod) unchanged
- [ ] Licence shows active with expected seat count

Technical and security

- [ ] Domain logon and service accounts work (AC-2)
- [ ] Only approved admin groups on the VM (AC-6)
- [ ] Configuration matches baseline (CM-2, CM-6)
- [ ] Backups configured and one completed (CP-9)
- [ ] Disks encrypted with managed keys (SC-12, SC-28)
- [ ] Audit and BarTender print logs collected (AU-2)

6.4 NIST SP 800-53 evidence

Control	What to show for this migration
AC-2, AC-6	Who held migration roles, for how long (PIM or Identity Center logs); local admin membership on the new VM.
CM-2, CM-3, CM-6	Both change records, the disk-freeze exception log, launch template or replication settings, post-migration configuration export.
CP-9, CP-10	Pre-migration backup, retained source VM as recovery path, cloud backup policy and first successful backup.
SC-12, SC-28	KMS key or disk encryption set used for replicated and final disks.
AU-2	CloudTrail or Azure Activity Log for every migration action; Windows and BarTender logs forwarded to the SIEM.

6.5 Rollback

A rehost has an unusually clean rollback, because the source VM is never modified: it is only shut down. As long as rollback starts before the decision deadline, the plant can be printing from the original server within minutes.

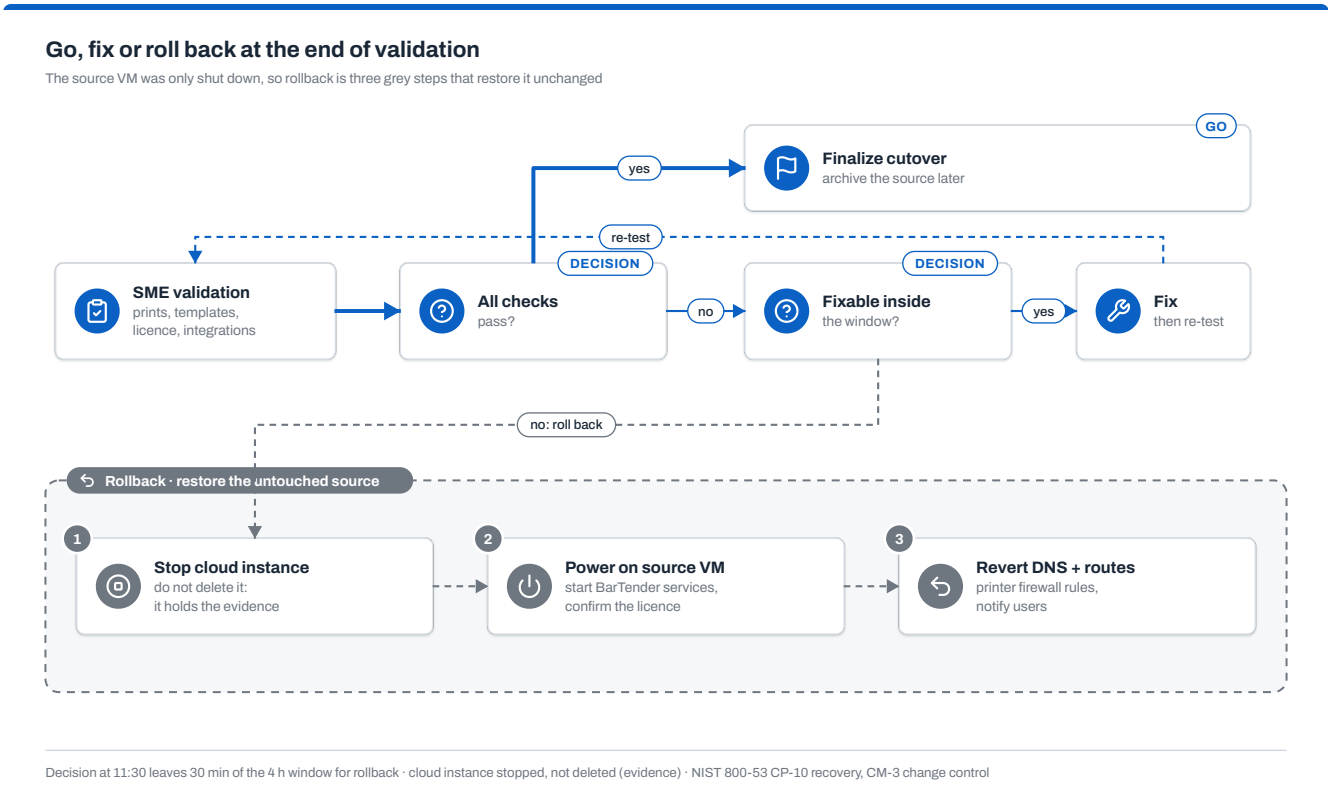


Figure 8. A rehost rolls back cleanly because the source VM is never modified, provided rollback starts before the decision deadline.

1. Stop the cloud instance (do not delete it; it holds evidence of what failed).
2. Power on the source VM; start BarTender services; confirm the licence (re-activate on the source if it was moved).
3. Revert the DNS record and any printer route or firewall change.
4. Notify users and record the failure reason. In MGN the server can be returned to *Ready for testing* and replication resumed; in Azure, replication continues until you remove it.

6.6 Troubleshooting

Symptom	Likely cause and fix
MGN server stuck in initial sync or "stalled"	TCP 1500 blocked from source to staging subnet, or 443 to MGN/S3 blocked. Check security groups, firewall and proxy.
Azure replication "Initial replication failed"	Appliance cannot reach ESXi hosts on TCP 902, missing vCenter privileges, or disk layout changed. Fix and re-enable replication.
Test instance does not boot	Driver or boot-mode issue. Check the serial console / boot diagnostics; confirm the OS is supported by the tool.
BarTender reports licence invalid	Activation bound to old hardware. Deactivate on source (or contact the vendor) and re-activate; open TCP 5160 if clients use a licence server.
Labels print slowly or time out	Latency over VPN, MTU issues, or printer status (SNMP) blocked. Measure RTT, check MTU/fragmentation, allow 161/UDP.
Some printers unreachable	Ports still use IP addresses on a range not routed from the cloud. Re-point to DNS names; add routes.
Templates or integrations fail to find files	Hard-coded UNC or drive letter. Use the discovery script output; update paths.
Dev/test instance off when needed	Off-hours schedule. Start manually or request a documented schedule exception.

CHAPTER 7

Summary, Glossary and Review

7.1 Chapter summary

- Rehost (lift and shift) is the default for validated label servers because block-level replication preserves the exact validated state.
- AWS MGN and Azure Migrate both replicate disks continuously or in delta cycles, allow repeated test launches, and cut over once inside a window.
- What changes is the network path, the licence binding and the hardware the OS sees; plan printers (TCP 9100), DNS and licensing explicitly.
- Keep the source VM untouched until finalization: it is the rollback.
- Rebuild (Method C) when the OS is unsupported or the business wants an upgrade, and expect fuller revalidation.

Check your understanding

1. Which of the 7 Rs describes Methods A and B, and why does it suit a validated labeling system?
2. What is the difference between a test launch and a cutover, and why can you repeat one but not the other?
3. Name three dependencies that commonly break when a BarTender server moves to the cloud, and the mitigation for each.
4. Which ports must be open for MGN replication, and for printing from the cloud to a thermal printer?
5. Describe the rollback path if SME validation fails 30 minutes before the end of the cutover window.

7.2 Glossary

Rehost	Moving a server unchanged onto cloud infrastructure ("lift and shift").
Block-level replication	Copying disk blocks below the file system, then sending only changed blocks.
AWS MGN	AWS Application Migration Service; agent-based rehost to EC2.
Azure Migrate	Azure hub for discovery, assessment and migration; Migration and modernization replicates servers to Azure VMs.
Cutover	The planned switch of production from source to cloud instance.
TCP 9100	Raw printing port used by most network label printers.